CSA5111 - Cryptography and Network security

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class test - 1

1) plain text = MEETATNOON

shift = 7

Enter key = SUN

Caesar cipher :-

Encryption :- $C = (p+K) \bmod 26$

$C = (p+7) \bmod 26$

0	1	2	3	4	5	6	7	8	9	10
A	B	C	D	E	F	G	H	I	J	K

11	12	13	14	15	16	17	18	19	20
L	M	N	O	P	Q	R	S	T	U

21	22	23	24	25
V	W	X	Y	Z

M $\Rightarrow (12+7) \bmod 26 \Rightarrow 19 \bmod 26 \Rightarrow 19 - T$

E $\Rightarrow (4+7) \bmod 26 = 11 \bmod 26 = 11 - L$

E $\Rightarrow L$

T $\Rightarrow (19+7) \bmod 26 = 26 \bmod 26 = 0 - A$

A $\Rightarrow (0+7) \bmod 26 = 7 \bmod 26 = 7 = H$

T $\Rightarrow A$

N $\Rightarrow (13+7) \bmod 26 \Rightarrow 20 \bmod 26 = U$

O ⇒ (14+7) mod 26 ⇒ 21 mod 26 = V

O ⇒ V

N ⇒ U

TKAH TLLAHAUVVU

decryption :- $p = (c-7) \bmod 26$

T ⇒ (19-7) mod 26 ⇒ 12 mod 26 ⇒ 12 ⇒ M

L ⇒ (11-7) mod 26 ⇒ 4 mod 26 ⇒ 4 ⇒ E

L ⇒ E

A ⇒ (0-7) mod 26 ⇒ -7 mod 26 ⇒ 19 ⇒ T

H ⇒ (7-7) mod 26 ⇒ 0 - A

A ⇒ (8-7) mod 26 ⇒ -7 mod 26 = 19 ⇒ T

U ⇒ (20-7) mod 26 ⇒ 13 mod 26 = N

V ⇒ (21-7) mod 26 ⇒ 14 mod 26 = O

V ⇒ O

U = N

P = MEETATNOON

Vigenere cipher

M E E T A T N O O N

S U N S U N S U N S

Encryption $C = (p+K) \bmod 26$

M $\Rightarrow (12+18) \bmod 26 = 4 = E$

E $\Rightarrow (4+20) \bmod 26 = 24 = Y$

E $\Rightarrow (4+13) \bmod 26 = 17 = R$

T $\Rightarrow (19+18) \bmod 26 = 15 = P$ L

A $\Rightarrow (0+20) \bmod 26 = 20 = U$

T $\Rightarrow (19+13) \bmod 26 = 6 = G$

N $\Rightarrow (13+18) \bmod 26 = 5 = F$

O $\Rightarrow (14+20) \bmod 26 = 8 = I$

O $\Rightarrow (14+13) \bmod 26 = 25 = Z$

N $\Rightarrow (13+18) \bmod 26 = 5 = F$

E Y R L U G F I Z F

M A R P U O N S Z V

S U N S U N S U N S

$P = (C-K) \bmod 26$

$M \Rightarrow (4-18) \bmod 26 = 12 = M$

$Y \Rightarrow (24-20) \bmod 26 = 4 = E = U = E$

$R \Rightarrow (17-13) \bmod 26 = 4 = E = U = E$

$P \Rightarrow (15-18) \bmod 26 = 19 = T$

$U \Rightarrow (20-20) \bmod 26 = 0 = A$

$G \Rightarrow (6-13) \bmod 26 = 19 = T$

$F \Rightarrow (5-18) \bmod 26 = 13 = N$

T ⇒ (18-20) mod 26 = 24 = O

Z ⇒ (25-13) mod 26 = 14 = O

F ⇒ (5-18) mod 26 = 13 = N

Plain text = MEETATNOON

2) Hill cipher

Plain text :- MATH

$$\text{Key} = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

Encryption :- $C = KP \bmod 26$

MATH ⇒ $P_1 \begin{bmatrix} 12 \\ 0 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix} P_2$

MA ⇒ $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix} \bmod 26$

⇒ $\begin{bmatrix} 36 \\ 60 \end{bmatrix} \bmod 26$

⇒ $\begin{bmatrix} 10 \\ 8 \end{bmatrix}$

10 = K 8 = I = $\begin{bmatrix} K \\ I \end{bmatrix}$

TH $\Rightarrow \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix} \bmod 26$

$\Rightarrow \begin{bmatrix} 71 \\ 100 \end{bmatrix} \bmod 26$

$\Rightarrow \begin{bmatrix} 19 \\ 14 \end{bmatrix} \Rightarrow$

T = 19 14 = O

Decryption = $K^{-1}C \bmod 26$

$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \Rightarrow \frac{1}{21-10} \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$

$K^{-1} = \frac{1}{11} \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \Rightarrow K^{-1} = 11^{-1} \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$

$K = \begin{bmatrix} 2 & 3 \\ -4 & 5 \end{bmatrix}$

$K \neq \text{Gcd}(14, 26) = 2$

So, It is unacceptable to calculate detesminant.

3) a) AES-256-GCM :-

* A secret key is used for both encryption and decryption.
* The sender and receiver uses same secret key.

RSA :-

* It consists public key and private key
* public key is used for encryption
* Private key is used for decryption.

b) * AES-256-GCM encrypts better than RSA
* the maximum data size per operation consists for RSA
* The data size of AES is 256 bits so, it is less efficient.

c) Scalability :-

No. of clinics = 200

200 x 1 = 200 shared keys.

The clinics most communicate pair wise versus RSA

$$\frac{n(n-1)}{2}$$

$\frac{200(200-1)}{2} \Rightarrow \frac{200(199)}{2} \Rightarrow 19900$ keys

4) $p = 11 \quad q = 19 \quad e = 7 \quad M = 5$

$n = p \times q$

$n = 11 \times 19$

$n = 209$

$\phi(n) = (p-1)(q-1)$

$= (11-1)(19-1)$

$= 10(18)$

$= 180$

$\gcd(e, \phi(n)) = 1$

$\gcd(7, 180) = 1$

$1 \bmod 180 = 7d$

$d = 103$

Encryption :- $c = M^e \bmod n$

$c = 5^7 \bmod 209$

$c = 168$

decryption :- $M = c^d \bmod n$

$M = 168^{103} \bmod n$

$M = 5$

5) Techniques :-

1) caeser cipher
2) playfair cipher
3) vigenere cipher

Caeser cipher :-

Encrypts and decrypts the data by replacing the letter by fixed number of position.

plain text : HELLO

Shift :- 3

$$C = (p + K) \bmod 26$$

H $\Rightarrow (7 + 3) \bmod 26 = 10 = K$

E $\Rightarrow (4 + 3) \bmod 26 = 7 = H$

L $\Rightarrow (11 + 3) \bmod 26 = 14 = O$

O $\Rightarrow (14 + 3) \bmod 26 = 17 = R$

KHOOR

* As well as replace the letter using decryption $p = (C - K) \bmod 26$

playfair cipher :-

Replacing the two letter at a time in encryption and decryption, by 5x5 key matrix

Key: MONARCHY

plain text: HELLO

divide plain text into two letters

HE | LX | LO

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

CFSUPM

⇒ As well as decryption replaces letter using 5x5 key matrix.

Vigenere cipher :- Encrypts data by repeating the key.

plain text = hello

Keyword = key

H	E	L	L	O
K	E	Y	K	E

Encryption :- $c = (p + k) \bmod 26$

H ⇒ $(7 + 10) \bmod 26 = 17 \bmod 26 = R$

E ⇒ $(4 + 4) \bmod 26 = 8 \bmod 26 = I$

L ⇒ $(11 + 24) \bmod 26 = 35 \bmod 26 = 9 = J$

L ⇒ $(11 + 10) \bmod 26 = 21 \bmod 26 = V$

$O \Rightarrow (14+4) \bmod 26 \Rightarrow 18 \bmod 26 \Rightarrow S$

$C = RIJVS$

⇒ decryption repeats same as encryption using formula $P = (C - K) \bmod 26$